

LING 313.02 – Problem Session 1

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Week 1 – Concepts

Phonetics vs. Phonology

Phonetics is the scientific study of the physical characteristics of speech sounds. It is concerned with how sounds are produced (articulatory phonetics), their physical properties like frequency and duration (acoustic phonetics), and how they are perceived (auditory phonetics). Our focus is on articulatory phonetics.

Phonology studies how sounds function and pattern within a specific language. It investigates which sound differences are meaningful (phonemes) and the rules governing how sounds can be combined (phonotactics).

Air-stream mechanisms All Turkish sounds are produced by a pulmonic egressive mechanism. Air is initiated and forced out by the lungs, then modified in the vocal tract. This is the most common air-stream mechanism across languages.

The velum and nasality The velum (soft palate) controls the nasal–oral distinction:

- When the velum is raised, it blocks the nasal cavity, producing oral sounds (e.g., [p], [t], [k], [s]).
- When the velum is lowered, air flows through the nasal cavity, producing nasal sounds (e.g., [m], [n]).

[m] and [b] are both voiced and bilabial. They differ in nasality: the velum is lowered for [m] and raised for [b].

The International Phonetic Alphabet (IPA)

Turkish orthography is largely phonetic, but it does not capture every distinction used in phonological analysis. The IPA gives a one-to-one correspondence between symbols and sounds.

- (1) *kar* ‘snow’ [k^har] vs. *kâr* ‘profit’ [c^har]
- (2) *sol* ‘left’ vs. *sol* ‘G note’ (the latter having a palatalized /l/)
- (3) *kuş* [kuʃ] (broader) vs. [k^huʃ] (narrower; ^h marks aspiration)

A broader transcription records less phonetic detail. A narrower one can show aspiration explicitly. The superscript ^h is a diacritic, not an extra segment.

Turkish vowels

Turkish has eight vowel phonemes, distinguished by height, backness, and rounding.

Height	Front		Back	
	Unrounded	Rounded	Unrounded	Rounded
High	<i>i</i>	<i>ü</i>	<i>ɯ</i>	<i>u</i>
Mid	<i>e</i>	<i>ö</i>	—	<i>o</i>
Low	—	—	<i>a</i>	—

ɯ and *i* are both high and unrounded. They differ in backness: *ɯ* is back and *i* is front. Orthographic *ı* corresponds to IPA [ɯ].

Week 2 – Phonemes, Allophones & Voiceless Stops

Summary

- **Phoneme:** a contrastive sound unit that can distinguish meaning. Minimal pairs differ by one sound in the same position and have different meanings. *at* ‘horse’ vs. *ad* ‘name’ supports /t/ vs. /d/. *kas* ‘muscle’ vs. *kaz* ‘goose’, and *sil* ‘erase’ vs. *zil* ‘bell’, test voicing.
- **Allophone:** a phonetic variant of a phoneme. Allophones of one phoneme do not distinguish meaning. Aspirated [t^h] and unaspirated [t] are both variants of /t/.
- **Complementary distribution:** two sounds never appear in the same environment. In the course model, unaspirated voiceless stops occur syllable-finally within a word, and aspirated ones occur elsewhere.
- A phonetic difference alone does not establish a phonemic contrast. Not yet finding a minimal pair is not proof that two sounds are allophones.

The voiceless stops of Turkish

Turkish has three voiceless stop phonemes: /p/, /t/, and /k/. Each has aspirated and unaspirated allophones. Aspiration is predictable in this model, so it is not a distinctive feature here.

- /p/ → voiceless bilabial stop
 - [p] (unaspirated) syllable-finally within a word (– \$). Ex.: *kapla* [ˈkap.la]
 - [p^h] (aspirated) elsewhere, including word-finally. Ex.: *pat*, *kapak* [kaˈp^hak], *top* [t^hop^h]
- /t/ → voiceless dental stop
 - [t] (unaspirated) syllable-finally within a word. Ex.: *atla* [ˈat.la], *etki* [ˈet.ki]
 - [t^h] (aspirated) elsewhere. Ex.: *taş* [t^haʃ], *ata* [aˈt^ha], *at* [at^h]
- /k/ → voiceless velar or palatal stop
 - Velar with back vowels: [k^h] syllable-initially, [k] syllable-finally within a word. Ex.: *kalem* [k^haˈlem], *kol* [k^hol]

- Palatal with front vowels: [c^h] syllable-initially, [c] syllable-finally within a word. Ex.: *kel* [c^hel], *ekmek* [ˈec.mec^h]

IPA [ç] is a voiceless palatal stop. It is not the sound spelled with Turkish *c*.

Palatal /c/ is not always an allophone of /k/

Predictable palatal realizations near front vowels are a separate fact from contrast. *kâr* has [c^h] before the back vowel [ɑ], so a “palatal only next to front vowels” claim does not cover every case.

- (4) *kar* ‘snow’ [k^har] vs. *kâr* ‘profit’ [c^har]

In this pair, aspiration and the rest of the environment are held constant. The initial stop differs in place and the meanings differ, so /k/ and /c/ can be analyzed as distinct phonemes in this environment.

Rule notation used so far

/X/ → [Y] / environment means “/X/ is realized as [Y] in that environment.”

- \$ = syllable boundary
- -- = the position of the target sound
 1. /t/ → [t] / -- \$ within a word; [t^h] in the other positions given in class
 2. /k/ → [c] / before a front vowel, in the regular examples; [k] with a back vowel

Place and aspiration are separate conditions. For *kol*, back vowel and word-initial position give [k^h]. For the first consonant of *ek.mek*, a front vowel and syllable-final position within the word give [c].

Let’s play Kahoot!

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